
An AI-Driven Multi-Vector Energy Framework for Korea's West Coast Corridor: Optimizing Grid Topologies via Multi-Agent Collaboration with PyPSA-Korea and Spatial-Text Fusion

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Abstract

South Korea's energy transition currently faces a structural deadlock known as the "Renewable Energy Paradox," where the resource-rich Honam region is projected to exceed 21.5 GW of renewable capacity by 2030, leading to massive curtailment, while the industrial Seoul Metropolitan Area (SMA) suffers from chronic power shortages. While the government's "West Coast Energy Highway" (HVDC) project aims to bridge this gap, this study identifies two critical risks in the single-linear infrastructure approach: the Time-Gap Risk (a decade-long infrastructure void starting in 2026) and the Social Conflict Risk (NIMBY resistance). We propose the AMPEB (AI-Enhanced Multi-Path Energy Bridge) framework, which disperses surplus power across four vectors: (1) Grid Reinforcement, (2) Storage (LAES), (3) Conversion (P2X), and (4) Demand Shift. Utilizing a "Dual-Engine AI Co-Scientist" system, we integrated PyPSA-Korea for techno-economic optimization with a novel Spatial-Text Fusion Algorithm to quantify social acceptance. Simulation results demonstrate that the integrated AMPEB scenario (S4) achieves a renewable acceptance rate of 103.1% and an LCOE of 92.1 KRW/kWh, significantly outperforming the grid-only expansion strategy. Notably, a Human-in-the-Loop (HITL) intervention—correcting the AI's initial data center siting from Godeok to the Solaseado Distributed Energy Special Zone—improved economic feasibility by 18%, validating the role of AI not just as a tool, but as a collaborative scientific partner.

1 Introduction

1.1 The Geopolitical and Structural Crisis of the Korean Grid I

As of 2026, the South Korean energy landscape is confronting a severe structural contradiction that threatens the nation's carbon neutrality goals and industrial competitiveness. This phenomenon, termed the "Renewable Energy Paradox," describes the spatial mismatch where solar and wind generation are heavily concentrated in the geographically favorable Honam region (Southwest), while power consumption remains centralized in the Seoul Metropolitan Area (SMA) and its surrounding industrial clusters.

The Honam region, comprising Gwangju, Jeonnam, and Jeonbuk, has become the epicenter of

Korea's renewable energy expansion due to its high solar irradiance and favorable wind conditions. According to the 11th Basic Plan for Electricity Supply and Demand, Honam's renewable capacity is projected to exceed 21.5 GW by 2030. However, the local peak demand in this region hovers around only 12 GW, creating a massive generation surplus during peak daylight hours. This surplus has resulted in frequent grid saturation events, forcing the Korea Electric Power Corporation (KEPCO) to implement curtailment orders that waste clean energy and undermine the economic viability of renewable projects. Without intervention, curtailment rates are expected to rise exponentially, potentially bankrupting small-scale solar operators who lack the financial resilience to withstand revenue volatility.

Conversely, the SMA faces a critical power shortage, exacerbated by the explosive demand from semiconductor clusters in Pyeongtaek and Yongin. These facilities require gigawatt-scale, high-quality power to maintain 24/7 operations. The inability to transmit Honam's clean energy to the SMA not only wastes resources but also forces the SMA to rely on expensive and carbon-intensive LNG peaking plants, creating a bifurcation in the national energy strategy.

1.2 The "West Coast Energy Highway" and Its Limitations

To resolve this bottleneck, the Ministry of Trade, Industry and Energy (MOTIE) has proposed the "West Coast Energy Highway," a massive infrastructure project costing approximately 11.5 trillion KRW. This project involves constructing an 8GW High-Voltage Direct Current (HVDC) submarine transmission line running along the west coast to transport power from Honam to the SMA. Major industrial players like LS Cable & System and LS ELECTRIC are deeply involved in localizing the HVDC technology, specifically Voltage-Source Converters (VSC), to facilitate this backbone.

While theoretically sound, our research identifies two critical risks inherent in relying solely on this single-linear infrastructure:

1. Time-Gap Risk: Current construction timelines indicate the HVDC system will not be fully operational until post-2036. This creates a dangerous "infrastructure void" of over a decade, starting in 2026, during which renewable curtailment in Honam will likely accelerate. The grid inertia issues associated with high renewable penetration cannot wait ten years for a solution.

2. Social Conflict Risk: As evidenced by the historical Miryang transmission tower disputes, linear infrastructure in Korea faces severe public opposition (NIMBY). The land acquisition and compensation processes for onshore converter stations are prone to indeterminate delays caused by local litigation and protests. The "Social Cost" of these delays is rarely factored into Levelized Cost of Energy (LCOE) calculations but frequently derails projects. The "Energy Highway" concept, while evocative of the Gyeongbu Expressway's economic impact, fails to account for the modern complexity of environmental and property rights activism.

1.3 The AMPEB Hypothesis: A Multi-Vector Approach

We propose the AMPEB (AI-Enhanced Multi-Path Energy Bridge) framework to transition the national strategy from a "single grid" reliance to a "multi-vector" ecosystem. AMPEB hypothesizes that grid stability and economic efficiency can be maximized by optimizing four distinct pathways simultaneously:

- **Path 1 (Grid):** Strategic reinforcement of existing AC lines and the planned HVDC corridor, optimized to minimize social friction.
- **Path 2 (Storage):** Deployment of Liquid Air Energy Storage (LAES) utilizing waste cold exergy from the Gwangyang LNG Terminal. Unlike Lithium-ion batteries, LAES poses zero fire risk and offers longer duration storage suitable for grid shifting.
- **Path 3 (Conversion):** Power-to-X (P2X) conversion, specifically Green Hydrogen and Ammonia production, to couple the power sector with industrial feedstock needs.
- **Path 4 (Demand Shift):** Relocation of energy-intensive data centers from the SMA to renewable generation sites in Honam, facilitated by "Demand Response" (DR) incentives and regulatory sandboxes under the Distributed Energy Activation Special Act.

2 Methodology: Dual-Engine AI Co-Scientist Framework

This study utilizes a "Dual-Engine" system where distinct AI agents handle physical and social constraints, integrated via a Human-in-the-Loop (HITL) protocol. This methodology represents a departure from traditional energy modeling by treating AI agents as specialized researchers—Co-Scientists—rather than simple calculators. The framework leverages the "PyPSA-Korea" open-source model as the physical engine and a Large Language Model (LLM) based Spatial-Text fusion algorithm as the social engine.

2.1 Engine 1: Techno-Economic Optimization (PyPSA-Korea)

AI Agent: GPT-4o (Code Generation & Optimization Strategy)

Tool: PyPSA-Korea (Open Source Energy System Model)

We employed PyPSA-Korea, a Python-based open-source energy modeling framework customized for the Korean grid topology. Developed to support South Korea's sustainable energy transition, PyPSA-Korea builds upon the PyPSA-Earth architecture to provide high-resolution investment and operational planning. The AI agent, GPT-4o, was tasked with architectural design and script generation to model the specific constraints of the West Coast corridor.

2.1.1 Network Topology and Mathematical Formulation

A simplified 2-Node Model was established to isolate the North-South bottleneck:

- **Node A (Honam):** Characterized by high generation capacity (PV/Wind), low local load, and the presence of existing LNG infrastructure (Gwangyang Terminal).
- **Node B (SMA):** Characterized by low generation capacity but extremely high load density (Data Centers/Semiconductors).

The optimization problem minimizes the total system cost, defined as:

$$\min \left(\sum_{n,s} c_{n,s} \cdot G_{n,s} + \sum_{n,s} o_{n,s} \cdot g_{n,s,t} + \sum_l c_l \cdot F_l + \text{Penalty}_{\text{curtail}} \right)$$

Where:

- $c_{n,s}$ is the annualized capital cost of generator/storage s at node n .
- $o_{n,s}$ is the variable operating cost.
- $g_{n,s,t}$ is the dispatch of unit s at time t .
- F_l is the capacity expansion of transmission line l .

2.1.2 LAES Thermodynamics Modeling

We modeled Liquid Air Energy Storage (LAES) not as a generic battery but as a thermodynamic cycle integrated with LNG regasification. Standard LAES systems suffer from low round-trip efficiency (~50%) due to the energy intensity of air liquefaction. However, by coupling the system with the Gwangyang LNG Terminal, we utilize the waste cold energy (-162°C) released during LNG regasification.

GPT-4o derived the modified efficiency parameters based on the following exergy recovery logic:

$$W_{\text{net}} = W_{\text{exp}} - (W_{\text{comp}} - W_{\text{cold_recovery}})$$

The AI agent configured the PyPSA storage unit with a Round-Trip Efficiency (RTE) of 65% (a 15% point increase over baseline) and a 30% reduction in CAPEX due to the removal of high-stage compressors, as the LNG cold stream provides pre-cooling.

2.2 Engine 2: Spatial-Text Fusion (Social Risk Analysis)

AI Agent: Claude 3.5 Sonnet (Unstructured Data Analysis)

Method: Spatial-Text Fusion Algorithm

To quantify the "Social Conflict Risk"—a variable typically absent in engineering studies—we developed a novel Social Conflict Index (SCI). Traditional LCOE models assume that if a transmission line is funded, it will be built. History in Korea proves otherwise; social opposition is a hard constraint that manifests as infinite delay. AI Agent: GPT-4o (Code Generation & Optimization Strategy).

2.2.1 Data Mining and Corpus Construction

We utilized the BigKinds database to aggregate 5,300 news articles and municipal complaints from 2020 to 2024 related to transmission infrastructure. The search query filtered for keywords such as "transmission tower" (송전탑), "electromagnetic waves" (전자파), "compensation" (보상), and "cancer clusters" (암 발생).

2.2.2 Sentiment Analysis and Index Formulation

Claude 3.5 Sonnet processed this corpus to extract sentiment polarity (S_{text}) and intensity. The AI was prompted to detect linguistic markers of "uncompromising opposition" (e.g., "fight to the death," "fundamental right to health").

The Social Conflict Index (SCI) for a given location i was formulated as:

$$SCI_i = \alpha(1 - S_{text}) + \beta \left(\frac{1}{D_{grid}} \right) + \gamma(H_{history})$$

Where:

- S_{text} is the normalized sentiment score (-1 to +1), where -1 is extreme opposition.
- D_{grid} is the proximity to dense residential areas (social impact radius).
- $H_{history}$ is a binary flag for regions with historical protest records (e.g., Miryang).
- α, β, γ are weighting factors.

This engine allowed us to assign a "Social Cost Penalty" to grid expansion paths that traversed high-conflict zones, effectively forcing the optimization model to seek alternative vectors like storage or demand shifting.

3 Simulation & Results

3.1 Scenario Definitions

We simulated four scenarios to test the AMPEB hypothesis against the baseline and the government's current plan:

- **S1 (Baseline):** Business as Usual. Reliance on AC grid reinforcement only. No major HVDC or storage expansion.
- **S2 (Grid Only):** Accelerated completion of West Coast HVDC (single vector). This mirrors the current "Energy Highway" plan.

- **S3 (Storage Integration):** HVDC + LAES deployment in Gwangyang. This introduces the temporal shifting vector.
- **S4 (AMPEB Integrated):** HVDC + LAES + P2X + Demand Shift (Data Center Relocation). This represents the full multi-vector ecosystem.

3.2 Quantitative Results

The integrated S4 scenario demonstrated superior performance across all critical metrics compared to the Grid-only scenario (S1) and the HVDC-centric scenario (S2).

Table 1: Comparative Simulation Metrics for 2030 Projection

Metric	S1 (Baseline)	S2 (HVDC Only)	S3 (Storage)	S4 (AMPEB)	Improvement (S4 vs S1)
LCOE (KRW/kWh)	145.0	128.5	115.2	92.1	-36.5%
RE Acceptance (%)	82.4%	94.1%	98.5%	103.1%	+20.7%p
Curtailment (GWh)	4,500	1,200	350	0	100% Reduction
Grid Congestion (hrs)	870	210	150	45	94.8% Reduction
Social Conflict (SCI)	0.85 (High)	0.62 (Med)	0.45 (Low)	0.18 (Very Low)	Significant Risk Reduction

- **Economic Viability:** S4 achieved an LCOE of 92.1 KRW/kWh. The cost reduction was driven primarily by two factors: (1) high arbitrage revenue from the LAES system utilizing free waste cold energy, and (2) the avoidance of massive grid congestion charges due to the localized consumption by relocated data centers.
- **Grid Stability:** S4 was the only scenario to achieve zero curtailment, effectively absorbing the 21.5 GW surplus in Honam. The "Acceptance Rate" exceeding 100% implies the system possesses enough flexibility to handle even higher renewable penetration than currently projected, essentially "future-proofing" the grid.

3.3 The "Human-in-the-Loop" (HITL) Innovation: The Solaseado Pivot

A critical scientific breakthrough occurred during the collaborative phase between the human researchers and the AI agents, highlighting the necessity of the "Co-Scientist" model. This incident validates that while AI serves as a powerful optimizer, human domain expertise is required to correct contextual blind spots.

- **The AI Blind Spot:** During the initial optimization of Path 4 (Demand Shift), GPT-4o recommended siting new data centers in Pyeongtaek Godeok. The AI's logic was heavily weighted by the social sentiment analysis from Engine 2, which showed positive public association with "Jobs," "Samsung," and "High-Tech Clusters" in that region. The AI inferred that "high acceptance" equals "optimal siting."
- **Human Intervention:** The human research team identified a critical flaw in this recommendation. While socially popular, Pyeongtaek is physically located in a grid-saturated node (Node B - SMA). Adding GW-scale data center load to Pyeongtaek would exacerbate the North-South transmission bottleneck, defeating the purpose of the study. Furthermore, the human team introduced a new policy variable: Article 43 of the Distributed Energy Activation Special Act, which provides incentives for power consumption in generation regions.
- **Correction & Re-Optimization:** The team instructed EXAONE 3.5, acting as the "Reviewer Agent," to re-evaluate sites based on a composite metric of "Grid Spare Capacity" + "Special Zone Incentives" + "Social Acceptance." The system pivoted the recommendation to Solaseado(Haenam). Solaseado is designated as a "Distributed Energy Special Zone," allowing for direct PPA (Power Purchase Agreement) capabilities and exemption from transmission grid fees.

- **Impact:** This human-guided correction reduced the final LCOE by an additional 18%. The Solaseado site benefited from zero transmission costs and lower land acquisition costs, factors the AI had initially undervalued compared to the brute-force sentiment score of Pyeongtaek. This effectively turned a "grid burden" (new load) into a "grid stabilizer" (local consumption of surplus RE).

4 Discussion

4.1 Policy Implications: From Highway to Ecosystem

The AMPEB framework confirms that the "West Coast Energy Highway" cannot be a standalone solution. The delays associated with linear infrastructure (Time-Gap Risk) and public opposition (Social Conflict Risk) are too great to rely on a single vector. The simulation proves that a multi-vector approach is not just a redundancy measure, but an economic imperative. Specifically, the "Demand Shift" vector—relocating data centers to places like Haenam Solaseado—provides the highest marginal benefit by deleting the need for transmission rather than just expanding it.

4.2 The Role of AI as a Co-Scientist

This study validates the premise of the "AI Co-Scientist." The AI agents did not merely perform calculations; they synthesized heterogeneous data—merging physical grid laws (Kirchhoff's laws via PyPSA) with unstructured social sentiment (BigKinds analysis). They identified trade-offs that human engineers might miss (e.g., the precise trade-off between LAES CAPEX and HVDC distance). Most importantly, the collaboration demonstrated that AI requires human contextual oversight to navigate complex regulatory landscapes, but once guided, can optimize systems far beyond human computational capacity.

5 Conclusion

We presented AMPEB, a framework that solves the Korean Renewable Energy Paradox by integrating grid, storage, conversion, and demand vectors. By utilizing a Dual-Engine AI system, we optimized this framework to achieve an LCOE of 92.1 KRW/kWh and a Social Conflict Index of 0.18. The successful HITL intervention regarding the Solaseado siting underscores the future of scientific research: a symbiotic relationship where AI provides computational breadth and humans provide contextual depth.

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